

2.1 The Periodic Table: Chemical Periodicity

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.1 The Periodic Table: Chemical Periodicity
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The labels had become detached from four bottles in the laboratory. A student realised that the contents of one of them could easily be identified because in the addition of water it would **not** give fumes of hydrogen chloride.

Which did **not** give the HCl fumes?

- A. AlCl_3
- B. SiCl_4
- C. PCl_5
- D. MgCl_2

[1 mark]

Question 2

What is the correct order of the decreasing melting points of the four chlorides shown?

HCl MgCl_2 CCl_4 S_2Cl_2

	Highest melting point → Lowest melting point			
A	HCl	S_2Cl_2	CCl_4	MgCl_2
B	HCl	CCl_4	S_2Cl_2	MgCl_2
C	MgCl_2	CCl_4	S_2Cl_2	HCl
D	CCl_4	S_2Cl_2	HCl	MgCl_2

[1 mark]

Question 3

Consecutive elements **X**, **Y** and **Z**, are in the third period of the Periodic Table. Element **Y** has the highest first ionisation energy and the lowest melting point of these three elements.

What could be the identities of **X**, **Y** and **Z**?

- A. Silicon, Phosphorus, Sulfur
- B. Sodium, Magnesium, Aluminium
- C. Aluminium, Silicon, Phosphorus
- D. Magnesium, Aluminium, Silicon

[1 mark]

Question 4

The period 4 elements gallium (Ga), germanium (Ge), arsenic (As) and selenium (Se) are the elements directly below aluminium, silicon, phosphorus and sulfur in the Periodic Table.

The properties of each period 4 element resemble those of the period 3 element directly above it.

Which period 4 elements form oxides that dissolve in water to give an acid solution?

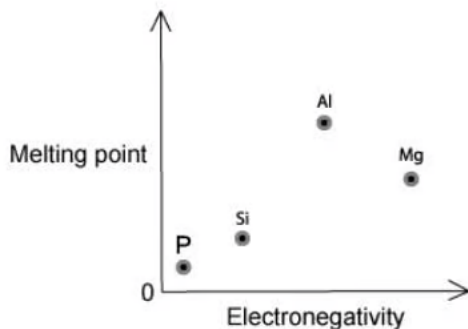
- A. Ga and Ge
- B. Ga and Se
- C. As and Se
- D. Se only

[1 mark]

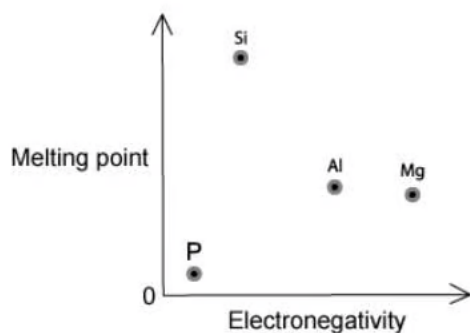
Question 5

Which graph correctly shows the relative melting points of period 3 elements plotted against their relative electronegativities?

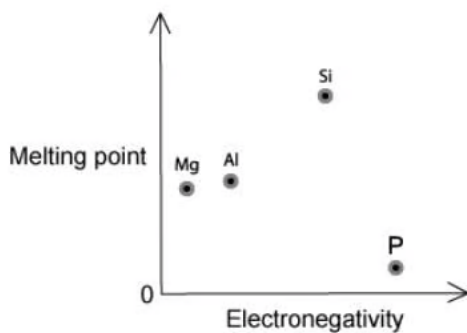
A.



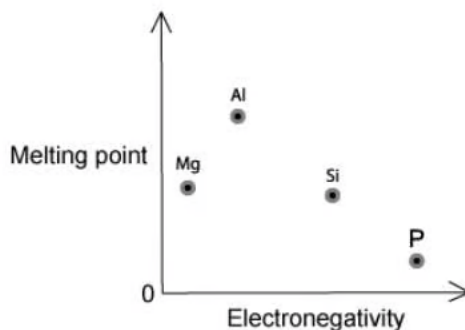
B.



C.



D.



[1 mark]

Question 6

A table of four substances and their physical properties is shown.

	Electrical conductivity of aqueous solution	Electrical conductivity of molten substance	Electrical conductivity of solid	Melting point / °C	Boiling point / °C
SiO ₂	Insoluble	Poor	Poor	1610	2230
NaCl	Insoluble	Good	Good	801	1465
Al	Good	Good	Good	100	340
SO ₃	Good	Poor	Poor	-72	-10

Which substances have the correct properties?

- A. SiO₂ and Al
- B. NaCl and Al
- C. SiO₂ and SO₃
- D. All

[1 mark]

Question 7

Non-metallic elements and their atoms can have the two following properties.

Property 1: Has **no** paired 3p electrons

Property 2: Has an oxide that can form a strong acid in water

Which properties do phosphorus and sulfur have?

	Sulfur	Phosphorus
A	2 only	1 and 2
B	1 and 2	2 only
C	1 and 2	1 and 2
D	2 only	1 only

[1 mark]

Question 8

Use of the periodic table is relevant to this question.

Elements **X** and **Y** react together to form compound **Z**. Elements **X** and **Y** are both in Period 3. Element **X** has the second smallest atomic radius in Period 3. Apart from argon, there is only one element in Period 3 which has a lower melting point than element **Y**.

Which compound could be **Z**?

- A. Na_2S
- B. MgS
- C. MgCl_2
- D. PCl_3

[1 mark]

Question 9

Which properties do compounds of silicon and aluminium have in common?

- A. Their oxides are amphoteric.
- B. Their oxides are insoluble in water.
- C. Aqueous solutions of their chlorides contain aluminium or silicon cations.
- D. Their chlorides have co-ordinate bonding.

[1 mark]**Question 10**

Period 3 elements can react with chlorine to produce the following chlorides:



Period 3 elements also react with oxygen to produce oxides.

When the oxides are produced is when the period 3 element is in its highest possible oxidation state. Which of the elements have a higher oxidation number than in the above chlorides?

- A. Mg, Al, Si, P and S
- B. Al, P and S
- C. Mg and Si
- D. P and S

[1 mark]

Model Answers: Medium

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The correct answer is **D** because:

- Magnesium chloride is an **ionic compound** and therefore dissociates into its ions (Mg^{2+} and Cl^-) and dissolves when added to water.
- **No reaction** takes place because neither ion has the ability to react with water, most importantly, the chloride ion is not able to take a hydrogen ion from water to make hydrochloric acid:
 - $\text{MgCl}_2(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{l}) + \text{Mg}^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq})$
 - Therefore, $\text{MgCl}_2(\text{s}) \rightarrow \text{MgCl}_2(\text{aq})$

A is incorrect as aluminium chloride is an **ionic compound** with a large amount of **covalent character**; it reacts readily with water to produce hydrochloric acid aluminium hydroxide: $\text{AlCl}_3 + 6\text{H}_2\text{O} \rightarrow [\text{Al}(\text{H}_2\text{O})_5\text{OH}]^{2+} + 3\text{HCl}$

B is incorrect as silicon chloride is a **covalent molecule** that **reacts readily** with water to produce hydrochloric acid and silicon oxide: $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$

C is incorrect as phosphorous chloride is a covalent molecule that reacts readily with water to produce hydrochloric acid and phosphoric acid: $\text{PCl}_5 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + \text{HCl}$

2

The correct answer is **C** because:

- Magnesium chloride has the **highest melting point** because it has a **giant ionic structure** and is **solid** at room temperature – lots of energy is required to break the strong ionic bonds to separate the molecules into a liquid state.
- CCl_4 and S_2Cl_2 are both **simple covalent molecules** with relatively **weak van der Waals' forces** between molecules – these do **not** require much energy to overcome, and therefore both molecules are **liquids** at room temperature.
- CCl_4 has a **larger molecular mass** than S_2Cl_2 and therefore has **stronger van der**

Waals' forces.

- HCl is a **simple ionic compound** which is a **gas** at room temperature because it has **weak van der Waals'** between molecules that requires little energy to overcome.

3

The correct answer is **A** because:

- First ionisation energies: silicon (X) < sulfur (Z) < phosphorous (Y)
- Melting points: phosphorous (Y) < sulfur (Z) < silicon (X)
- The **first ionisation energy** is the energy required to **remove one mole** of electrons (the easiest one to remove) from one mole of **gaseous atoms** to produce one mole of gaseous (positive) ions:
 - $\text{X(g)} \rightarrow \text{X}^+(\text{g}) + \text{e}^-$
- There is a **general increase** in first ionisation energy **across a period** because of the **nuclear charge** increases.
 - Therefore, so does the pull on the electron being removed – hence **more energy** is required to remove it.
- However, there is a reduction in first ionisation energy between groups 2 and 3 and between **groups 5 and 6**.
- Therefore, sulfur (Z) has lower first ionisation energy than phosphorus (Y) because the electron being removed is **paired**:

↑	↑	↑
↑↓	↑	↑

- Phosphorus 3p configuration:
- Sulfur 3p configuration:
- Silicon (X) has the highest **melting point** because it has a **giant covalent structure** so strong covalent bonds must be broken to melt it.
- Phosphorus (P_4) and Sulfur (S_8) are both **simple covalent molecules**, but sulfur has a higher melting point because it contains **more electrons**, so the **van der Waals'**

forces are stronger.

B is incorrect as First ionisation energies: $\text{Na} < \text{Al} < \text{Mg}$. Melting points: $\text{Na} < \text{Mg} < \text{Al}$.

C is incorrect as First ionisation energies: $\text{Al} < \text{Si} < \text{P}$. Melting points: $\text{P} < \text{Al} < \text{Si}$.

D is incorrect as First ionisation energies: $\text{Al} < \text{Mg} < \text{Si}$. Melting points: $\text{Mg} < \text{Al} < \text{Si}$.

4

The correct answer is **C** because:

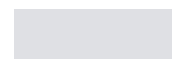
- In period 3 the following oxides produce an **acidic solution** when added to water:
 - Phosphorus oxide: $\text{P}_4\text{O}_6 + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_3$ **or** $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$
 - Sulfur dioxide: $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$
 - H_3PO_3 , H_3PO_4 and H_2SO_4 are all acids because they **dissociate** in water, **releasing H^+ ions**.
- Therefore, **arsenic** and **selenium oxides** will also produce an **acidic solution**.

A is incorrect as aluminium (and therefore Ga) oxide **does not react simply** with water because despite containing **oxide ions**, they are **held too strongly** in the solid lattice to react with the water. Silicon (and therefore Ge) oxide does not react with water because it has a giant covalent structure.

B is incorrect as silicon (and therefore Ge) oxide does not react with water because it has a giant covalent structure.

D is incorrect as phosphorous (and therefore As) oxides also react with water to produce an acidic solution.

5



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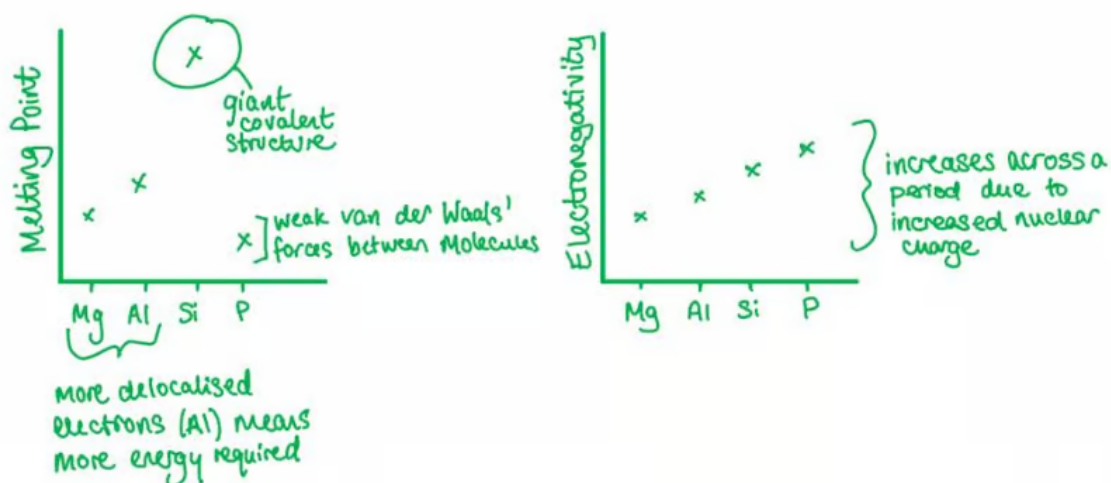
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The correct answer is **C** because:

Melting point:

- Silicon has the highest melting point because it is a **giant covalent structure**, so to turn silicon from a solid to a liquid you have to break many strong covalent bonds which require lots of **energy**.
- The melting point of the **metals** (Na, Mg, Al) increases across the period because the number of electrons **delocalised** from each metal atom increases (1 from Na, 2 from Mg, 3 from Al).
 - Therefore, more energy is required to force the delocalised electrons back into the electron shell.

Phosphorus and sulfur exist as **simple covalent molecules**, which means they have strong covalent bonds within molecules but only low amounts of energy are required to overcome the **van der Waals forces** between molecules in order to melt the solid.



Electronegativity:

- Electronegativity is a measure of the tendency of an atom to **attract a bonding pair** of electrons.
- Electronegativity **increases across a period** between group 1 and group 7.
- This is because the number of electron orbitals remains the same, but the **nuclear charge increases** (because the number of protons increases).
- This means that there is a **stronger nuclear pull** on the bonding electrons in the outer orbital.

6

The correct answer is **C** because:

- SiO₂** has a **giant covalent structure** and therefore has **strong covalent bonds** between molecules which give it its **high** melting and boiling point and **lack of solubility**.
- SO₃** is a **simple covalent molecule** that is a gas at room temperature and therefore has a **low** melting and boiling point, this is due to weak van der Waals' forces between molecules.
- Unlike SiO₂, SO₃ **dissolves** in water to produce **sulfuric acid**, H₂SO₄, this is a **strong acid** and therefore **completely dissociates** into its ions in solution – these ions are **mobile** and able to carry electrical charge and therefore **conduct electricity**.

- Both SiO_2 and SO_3 are covalent and therefore do **not** contain delocalised electrons and therefore **cannot** conduct electricity in the **molten or solid** form.

A is incorrect as **aluminium** metal has **relatively high** melting and boiling point because of the delocalised electrons requiring a lot of energy to return to the outer electron orbital.

B & D are incorrect as **NaCl** (table salt) is **soluble** in water, and it's solid does **not** conduct electricity because the **ions are bound** (immobile) within an ionic lattice and therefore **cannot carry a charge** and the melting and boiling points for aluminium are incorrect.

7

The correct answer is **D** because:

- Sulfur trioxide reacts with water to produce a **strong** acidic solution in water:
 - $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$
- A 'strong' acid is one that **fully dissociates** into its ions when in aqueous solution producing a high concentration of **H^+ ions** and therefore has a very low pH (1).
- The electronic configuration of phosphorus is $1s^2 2s^2 2p^6 3s^2 3p^3$.



- Due to **electron repulsion**, the 3 electrons in the 3p orbital are found separated across different subshells.

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- Due to **electron repulsion**, the 3 electrons in the 3p orbital are found separated across different subshells.

Exam hint: There is another common sulfur oxide – sulfur dioxide, SO_2 – which also produces acid when added to water: $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$. However, **H₂SO₃** is only a **weak acid** as it doesn't **fully dissociate** into ions in solution.

A is incorrect as phosphorus oxides only produce weak acids (H_3PO_3 and H_3PO_4) that do not fully dissociate in solution.

B & C are incorrect as the electronic configuration for sulfur is $1s^2 2s^2 2p^6 3s^2 3p^4$ and therefore has one electron pair in the 3p orbital phosphorus oxides only produce weak acids (H_3PO_3 and H_3PO_4) that do not fully dissociate in solution.

8

The correct answer is **D** because:

- Chlorine has the smallest **atomic radius** in period 3 (other than argon) because it has the largest number of **protons** and therefore the highest **nuclear charge**.
- A bigger nuclear charge means a greater **pull** on the electrons to the nucleus, reducing the atomic radius, so atomic radius **reduces** across a period.
- As argon is unreactive, **X** must be chlorine.
- Phosphorus (P_4) and sulfur (S_8) are **simple covalent molecules**, so the melting point is determined by the energy required to break **van der Waals'** forces between molecules.
- However, there are more electrons in sulfur molecules so phosphorus has a lower melting point than sulfur. Only chlorine and argon have lower melting points in Period 3.
- Therefore, the information is consistent with **Y** being **phosphorous** and **Z** being **PCl_3** .

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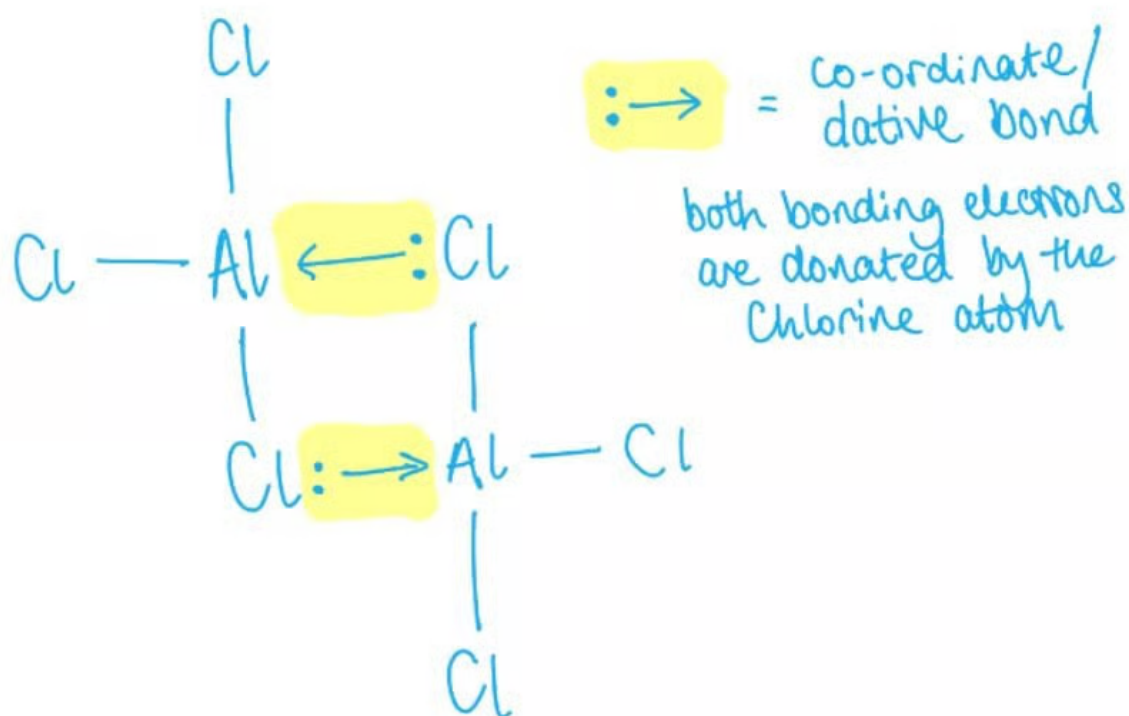
The correct answer is **B** because:

- **Aluminium oxide** does **not** react with water because the **oxide ion** is too **strongly attracted** to the aluminium ion.
- **Silicon oxide** has a **giant covalent structure**, and a **lot of energy** is required to overcome the strong covalent bonds between molecules so it will **not** dissolve.

A is incorrect as only **Al_2O_3** is **amphoteric** (acts as an acid and a base), **SiO_2** is **slightly acidic** and will react with strong bases.

C is incorrect as **neither** solution of their chlorides will contain aluminium or silicon cations because aluminium chloride reacts with water to form a **hydrated aluminium ion** **$[Al(H_2O)_6]^{3+}$** and silicon chloride is a **covalent molecule** so does **not** dissociate into ions.

D is incorrect as only aluminium chloride has co-ordinate (dative) bonding.



10

The correct answer is **D** because:

- The **oxidation states** of the period 3 elements in the chlorides in this question are:
 - $\text{MgCl}_2 = +2$ $\text{AlCl}_3 = +3$ $\text{SiCl}_4 = +4$ $\text{PCl}_3 = +3$ $\text{SCl}_2 = +2$
- The oxidation states of the period 3 elements in their **highest possible oxides** oxidation states are:
 - $\text{MgO} = +2$ $\text{Al}_2\text{O}_3 = +3$ $\text{SiO}_2 = +4$ **$\text{P}_4\text{O}_{10} = +5$** **$\text{SO}_3 = +6$**
- Therefore, the oxidation state of phosphorus and sulfur are **higher** in their oxides than the chlorides